```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
%matplotlib inline
```

```
from google.colab import files
f = files.upload()
```

Choose Files country_pro…ariables.csv

- **country_profile_variables.csv**(text/csv) - 9104 bytes, last modified: 10/29/2019 - 100% done

```
Saving country_profile_variables.csv to country_profile_variables.csv
```

```
# Step# 1
#

country = pd.read_csv('country_profile_variables.csv')
print(country.head(5))
print(country.tail(5))
print(country.shape)

print(type(country))
```

```
   Unnamed: 0         Country  Surface area (km2)  Population 1,000  (2017)  \
0           1     Afghanistan              652864                     35530
1           2         Albania               28748                      2930
2           3         Algeria             2381741                     41318
3           4  American Samoa                 199                        56
4           5          Angola             1246700                     29784

   GDP (M US$)  Life Exp Female  Life Exp Male
0        20270             63.5           61.0
1        11541             79.9           75.6
2       164779             76.5           74.1
3          -99             77.8           71.1
4       117955             63.0           57.4
     Unnamed: 0                    Country  Surface area (km2)  \
206         207  Wallis and Futuna Islands                 142
207         208             Western Sahara              266000
208         209                      Yemen              527968
209         210                     Zambia              752612
210         211                   Zimbabwe              390757

     Population 1,000  (2017)  GDP (M US$)  Life Exp Female  Life Exp Male
206                        12          -99             78.7           72.8
207                       553          -99             70.3           66.9
208                     28250        29688             65.6           62.8
209                     17094        21255             61.9           57.5
210                     16530        13893             59.0           56.1
(211, 7)
<class 'pandas.core.frame.DataFrame'>
```

```
lifeExpFemale = country['Life Exp Female']
lifeExpMale = country['Life Exp Male']

diffLE_MaleFemale = lifeExpFemale - lifeExpMale
averageLifeExp = (lifeExpFemale + lifeExpMale)/2
print(averageLifeExp)
```

```
0      62.25
1      77.75
2      75.30
3      74.45
4      60.20
       ...
206    75.75
207    68.60
208    64.20
209    59.70
210    57.55
Length: 211, dtype: float64
```

```
country['averageLifeExp'] = averageLifeExp
country['Difference'] = diffLE_MaleFemale
```

```
print(country.head(5))
```

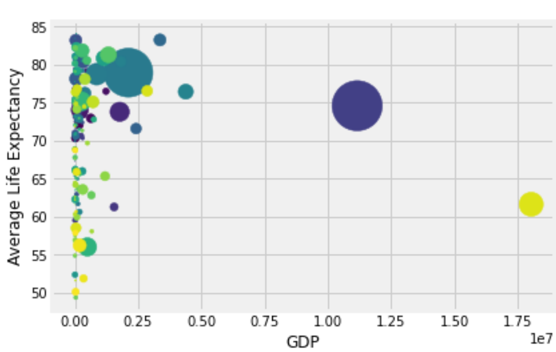
```
   Unnamed: 0         Country  Surface area (km2)  Population 1,000  (2017)  \
0           1     Afghanistan              652864                    35530
1           2         Albania               28748                     2930
2           3         Algeria             2381741                    41318
3           4  American Samoa                 199                       56
4           5          Angola             1246700                    29784

   GDP (M US$)  Life Exp Female  Life Exp Male  averageLifeExp  Difference
0        20270             63.5           61.0           62.25         2.5
1        11541             79.9           75.6           77.75         4.3
2       164779             76.5           74.1           75.30         2.4
3          -99             77.8           71.1           74.45         6.7
4       117955             63.0           57.4           60.20         5.6
```

```
sortedTable = country.sort_values(['Difference'],ascending=False)
print(sortedTable.head(5))
print(sortedTable.shape)
sortedTable.loc[sortedTable['Country'] ==  'United States of America']
```

```
     Unnamed: 0              Country  Surface area (km2)  \
182         183  Syrian Arab Republic              185180
154         155    Russian Federation            17098246
16           17               Belarus              207600
107         108             Lithuania               65286
102         103                Latvia               64573

     Population 1,000  (2017)  GDP (M US$)  Life Exp Female  Life Exp Male  \
182                     18270        28393             76.3           64.4
154                    143990      1326016             75.9           64.7
16                       9468        54609             77.7           66.5
107                      2890        41402             79.3           68.5
102                      1950        27004             78.7           68.8

     averageLifeExp  Difference
182           70.35        11.9
154           70.30        11.2
16            72.10        11.2
107           73.90        10.8
102           73.75         9.9
(211, 9)
```

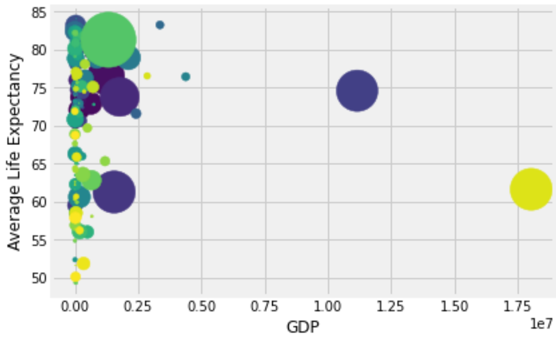
Unnamed: 0	Country	Surface area (km2)	Population 1,000 (2017)	GDP (M US$)	Life Exp Female	Life Exp Male	averageLifeExp	Differer
	United							

```
countryName = country['Country']
GDP = country['GDP (M US$)']
averageLifeExp = sortedTable['averageLifeExp']

marker_size_P = country['Population 1,000  (2017)']/1000 # Proportional to the Population
marker_size_SA = country['Surface area (km2)']/10000 # Proportional to the Surface Area



fig, ax = plt.subplots()

ax.scatter(GDP, averageLifeExp, s=marker_size_P, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```



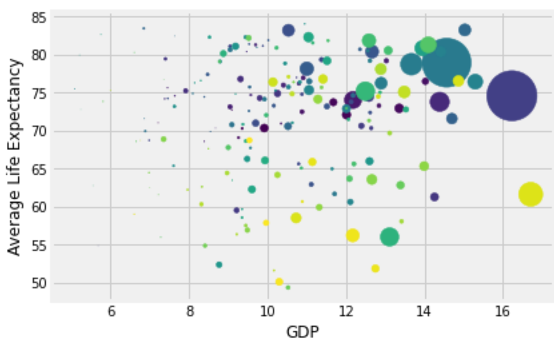
```
fig, ax = plt.subplots()

ax.scatter(GDP, averageLifeExp, s=marker_size_SA, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```

```
/usr/local/lib/python3.9/dist-packages/matplotlib/collections.py:963: RuntimeWarning: invali
  scale = np.sqrt(self._sizes) * dpi / 72.0 * self._factor
```

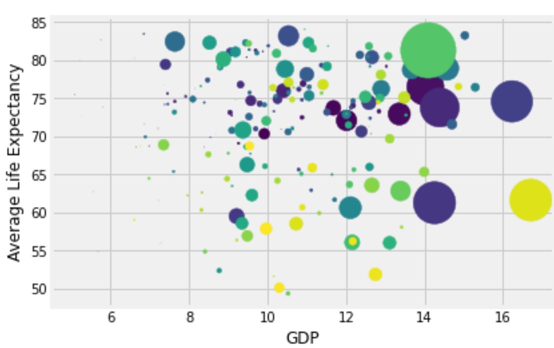



```
fig, ax = plt.subplots()

ax.scatter(np.log(GDP), averageLifeExp, s=marker_size_P, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```

```
/usr/local/lib/python3.9/dist-packages/pandas/core/arraylike.py:397: RuntimeWarning: invalid
  result = getattr(ufunc, method)(*inputs, **kwargs)
```




```
fig, ax = plt.subplots()

ax.scatter(np.log(GDP), averageLifeExp, s=marker_size_SA, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```



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